

CIP-BTSB-BT-001 Bench Test Final Report

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BENCH TEST — FINAL REPORT

Latent-heat thermal buffering across a 3 mm inter-plate layer

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[PRE-GEN-0 MECHANISM DEMONSTRATION — NOT A PRODUCT VALIDATION]

This report describes a bench-scale test of a single physical mechanism. It tested one layer of a five-layer design, using a substitute material, with no battery cell present and no thermal runaway event. **It says nothing about BTSB as a product.**

1. SUMMARY

A three-condition bench test was performed to determine whether a phase-change material layer produces a **measurable** delay and absorption of heat crossing between two surfaces — the mechanism underlying Layer 2 of the BTSB design.

The protocol, predictions, analysis methods and falsification criteria were published to a public repository with a PGP-verified timestamp **before any heated run was performed**.

RUN C	plates in direct contact	t85 = 22.04 min
RUN B	3 mm liquid paraffin (no phase change)	t85 = 37.42 min
RUN A	3 mm paraffin wax (phase change)	t85 > 45 min

Three independent lines of evidence, derived by different methods from one dataset, support the same conclusion:

1. A 14-minute suppression of the hot-plate climb rate, occurring inside both the predicted duration band (13–25 min) and the predicted temperature band (58–68 °C), with no corresponding feature in the control.
2. An energy balance in which approximately 3,900 J entered the fixture and did not appear as plate temperature – 85 % of the theoretical heat of fusion of the charge – ceasing to accumulate at the same moment the rate plateau ends.
3. The pre-specified power-measurement method, applied without modification, reporting 36 % of the baseline delivered power.

Latent-heat thermal buffering is demonstrated quantitatively.

Of seven pre-registered quantitative predictions, **five failed and two passed**. The two that passed are the two that test the mechanism. The five that failed are all predictions about the apparatus. Both outcomes are reported.

Total programme cost: under ₹8,000. No laboratory, no institutional support, one operator.

2. SCOPE

2.1 What this test establishes

That a 3 mm layer of phase-change material placed between two aluminium plates, heated at constant power, absorbs a quantity of energy consistent with its heat of fusion, and suppresses the temperature rise of the heated plate for a period consistent with prediction.

2.2 What this test does not establish

- x Nothing about BTSB as an assembled five-layer product
- x Nothing about performance against a real cell in thermal runaway
- x Nothing about propagation between cells
- x Nothing about the four other layers – pyrolytic graphite, expandable graphite, zeolite/carbon gas capture, PVDF dielectric
- x Nothing about RT64HC, the specified PCM. A paraffin wax substitute was used.
- x Nothing about durability, ageing, moisture uptake, or cycling

3. PRE-REGISTRATION AND AUDIT TRAIL

The distinguishing feature of this programme is that **every prediction was published before the data that tests it existed**, and every subsequent decision was published before the runs it affects.

3.1 Governing commit

Repository	github.com/Idreesh111/btsb-bench-test-preregistration
Commit	b489017e6244784ee108d8f81bd0c35d30bdbb4c
Timestamp	2026-08-26 T 01:08:48 Z = 06:38:48 IST
Signature	PGP verified "verified": true, "reason": "valid"

The first heated run of the programme began after this commit. The commit contains the full protocol, the predicted values of §7.2, the analysis methods of §6, and the falsification criteria of §9.

3.2 Document sequence — each issued before the runs it governs

Document	Issued	Governs
Record G1 — software chain	25 Aug	all runs
Record G2 — sensing	24 Aug, rev. 25 Aug	all runs
Record G3 — heater circuit	25 Aug	all runs
CIP-BTSB-BT-001 Rev A	26 Aug, 06:38 IST	all runs
Record R-C — Run C + deviation	26 Aug	Runs B and A
Record R-B — Run B + lower-bound method	27 Aug	Run A
Record R-A — Run A	29 Aug	—
This report	29 Aug	consolidates the above

This report does not supersede those records. They remain the primary audit trail precisely because they were issued in sequence. A consolidated document written after all data exists cannot demonstrate what a dated sequence demonstrates.

3.3 Two decisions published before the data they affect

26 August — the termination limit. Run C reached its endpoint at 22.04 min against a predicted 10–16 min. Record R-C extended the §5.5 termination limit from 30 to 45 minutes for Runs B and A, published with its reason, **before either run was performed**. All safety limits were left untouched.

27 August — the lower-bound reporting method. Record R-B §8 stated that Run A would probably not reach 85 °C inside 45 minutes, and specified in advance how the result would be reported if it did not:

“The limit will not be extended a second time. It was set at 45 minutes on 26 August, published with its reason, before Runs B and A existed. Moving it again after seeing Run B’s timing would be the precise failure that pre-registration exists to prevent.”

Run A did not reach 85 °C. **The published method was applied without alteration.**

4. APPARATUS

4.1 Fixture

Plates	2 × 5052-H32 aluminium, 173 × 71 × 3 mm measured: H 98.24 g C 97.83 g pair 196.07 g C_plates = 196.07 g × 0.88 J/g·K = 172.5 J/K
Cavity	155 × 55 × 3 mm window in a silicone gasket volume 25.575 cm³
Clamping	4 × G-clamp at 43 / 60 / 105 / 130 mm finger tight + ½ turn
Insulation	ceramic fibre blanket, 1260 °C rating full enclosure, clamps wrapped, edges packed
Orientation	heated from above

4.2 Heating

3 × 12 V 40 W ceramic cartridge heaters, 6 × 20 mm, in SERIES
mounted on plate H at 43 / 86 / 130 mm across the 71 mm width

Measured resistance 10.17 Ω (series confirmed:
parallel would read 1.11 Ω)
Supply 11.8 V × 1.16 A = 13.7 W electrical

4.3 Instrumentation

Arduino Uno R3 · 2 × MAX6675 · 2 × K-type M6 screw probes
SCK D6 · S0 D5 · CS1 D7 (TC1, hot plate) · CS2 D8 (TC2, cold plate)
1 Hz CSV logging · measured interval 1.003 s (+0.26 %)
Resolution 0.25 °C · K-type Class 2 absolute tolerance ±2.5 °C

5. VERIFICATION GATES

All three gates were cleared before the first heated run. Full detail in Records G1, G2 and G3.

Gate	Verified	Result
G1 — software chain	sketch compiles, logs, sampling interval measured	PASSED
G2 — sensing	836 readings, zero faults; 0.25 °C resolution confirmed; common-mode drift rejected by ~45×; TC2 verified at 101.0 °C against a 100.0 °C boiling reference; TC1 chained to TC2 at 93–95 °C	PASSED
G3 — heater circuit	series topology confirmed by resistance; delivered power measured	PASSED

Sensor-fault rate across the entire programme, 7,457 logged samples: 0.00 %.

5.1 One offset instruction was issued and withdrawn

Record G2 initially interpreted a −0.63 °C still-air channel difference as an instrument offset and directed that +0.63 °C be added to every measured difference.

The boiling-water checks included an interval with **both probe tips held physically together in air**, which read **−0.03 °C** — twenty times smaller. The still-air figure had contained an unknown mixture of instrument offset and genuine spatial temperature difference in the room, and the two could not be separated from that dataset.

The instruction was withdrawn in full and was never applied to any run. It was replaced by a per-run baseline measured in the as-built configuration immediately before each run. Had it not been caught, a fixed offset applied consistently across all three runs would have displaced every reported temperature difference in the same direction and looked entirely self-consistent in the results.

6. METHOD

Three conditions, identical in every respect except the layer between the plates.

RUN C	plates in direct contact through a thin thermal-paste film no gasket, no layer BASELINE
RUN B	3 mm liquid paraffin (mineral oil), 25.6 mL chemically similar to wax; does NOT change phase in the test range CONTROL
RUN A	3 mm paraffin wax, 23.00 g, cast solid DOES change phase in the test range TEST

C → B	isolates the delay caused by A LAYER BEING PRESENT
B → A	isolates the delay caused by MELTING

Run B is what makes Run A interpretable. Without it, any delay in Run A could be attributed to thermal resistance rather than latent heat.

6.1 Primary endpoint

t₈₅ — elapsed time from power-on until TC2 (cold plate) reaches 85.0 °C, by linear interpolation between straddling samples.

6.2 Delivered power — pre-specified

Linear least-squares fit to the mean of TC1 and TC2 over the interval in which that mean lies between 40 and 70 °C, converted to power by C_{plates} = 172.5 J/K.

6.3 Hypothesis and falsification

H1 A phase-change layer delays heat transfer measurably more than a non-phase-change layer of identical thickness.

H1 IS FALSIFIED IF $\Delta t_{\text{latent}} = t_{85}(\text{A}) - t_{85}(\text{B}) \leq 60 \text{ s}$

7. DEVIATIONS AND APPARATUS CHANGES

Every change to the apparatus or protocol during the programme, in order.

7.1 Run C-1 aborted — insulation inadequate (26 Aug)

The first heated run was terminated at 12.4 minutes. A heat-loss model fitted to 700 s of data gave $P = 9.07 \text{ W}$, $k = 0.221 \text{ W/K}$, and an equilibrium ceiling of approximately **76 °C** — below the 85 °C endpoint. The fixture was shedding heat as fast as the heaters supplied it.

The four G-clamps were the dominant loss path: steel in direct contact with a heated plate, projecting into ambient air, functioning as cooling fins.

Insulation rebuilt — clamps wrapped, blanket pressed onto the heaters, edges packed

loss coefficient	0.221 → 0.106 W/K	52 % reduction
TC2 ceiling	~76 → ~126 °C	

No change to heaters, supply, plates, sensors, clamping or firmware. Run C-1's raw data is published as

`run_C_ABORTED_20260826.csv`.

7.2 Termination limit extended, 30 → 45 min (26 Aug)

Published in Record R-C **before Runs B and A**. Safety limits unchanged: TC1 = 110 °C early stop, TC1 = 120 °C hard abort, all abort criteria, the primary endpoint, the analysis methods, the falsification criteria and the predicted values all untouched.

7.3 Aluminium foil liner (27 Aug)

The first attempt to fill the cavity for Run B leaked continuously at room temperature. Degreasing and re-clamping failed. **An 11 µm aluminium foil liner** was pressed into the gasket window and folded over the rim, forming a continuous pocket.

Alternatives rejected: external tape (adhesive does not bond to oil; PVC softens at 80 °C); RTV silicone sealant (24 h cure, and a hand-laid bead changes gasket thickness by an unrepeatable amount, making cavity volume differ between Runs B and A).

The foil creates a thermal bypass, identified during Run B and quantified:

Window perimeter	0.420 m
Foil thickness	11 µm
Bypass cross-section	$4.62 \times 10^{-6} \text{ m}^2$
k (aluminium)	200 W/m·K
Path length	3 mm

OIL layer conductance	0.398 W/K
FOIL bypass	0.308 W/K

The foil carries 44 % of the heat crossing the layer.

The same liner was used in Run A. The bypass is present in both runs being compared, and it biases both $\Delta t_{\text{resistance}}$ and Δt_{latent} **downward** — against the mechanism, not in favour of it. The figures in this report are conservative.

7.4 Cast-wax contact resistance — unanticipated (29 Aug)

Not a deliberate change; an unanticipated property of the Run A configuration. See §9.3.

8. RESULTS

8.1 Consolidated

	RUN C-2	RUN B	RUN A
Layer	direct contact	liquid paraffin	paraffin wax 23.00 g
Samples	1,600	2,501	3,356
Sensor faults	0	0	0
Ambient at start (°C)	36.17	38.09	37.23
Baseline TC1 – TC2 (°C)	–0.043	–0.289	–0.400
Heating duration (min)	22.04	37.42	45.06
t₈₅ (min)	22.04	37.42	> 45
Peak TC1 (°C)	88.00	92.25	89.75
Peak TC2 (°C)	86.75	86.25	80.25
TC1 – TC2 steady (°C)	2.19 ± 0.41	7.38 ± 0.49	9.75 ± 1.26
Delivered power \$6.2 (W)	7.73	5.42	2.77

Baseline statistics are computed over all samples before power-on. This convention differs by less than 0.02 °C from the windows used in the individual records; the published record values remain the governing figures and the two agree.

All three runs started within the ± 2 °C band required by §5.3. Peak temperatures occur **after** power-off in every run, consistent with the fixture’s thermal lag.

8.2 Resistance delay — measured

$$\begin{aligned}
 t_{85} \text{ (Run C)} &= 22.04 \text{ min} \\
 t_{85} \text{ (Run B)} &= 37.42 \text{ min} \\
 \hline
 \Delta t_{\text{resistance}} &= 923 \text{ s} = 15.38 \text{ min}
 \end{aligned}$$

This is the delay produced by a 3 mm non-melting layer.

8.3 Latent-heat delay — lower bound

Run A did not reach 85 °C within the 45-minute limit. TC2 peaked at 80.25 °C.

$$\begin{aligned}
 t_{85} \text{ (Run A)} &> 45.0 \text{ min} \\
 t_{85} \text{ (Run B)} &= 37.42 \text{ min} \\
 \hline
 \Delta t_{\text{latent}} &> 455 \text{ s}
 \end{aligned}$$

H1 is falsified if $\Delta t_{\text{latent}} \leq 60 \text{ s}$.
The lower bound exceeds that threshold by a factor of 7.6.
H1 IS NOT FALSIFIED.

9. ANALYSIS

9.1 The plateau

TC1’s climb rate in two-minute windows, Run A against Run B:

minutes	RUN A	RUN B	
8 – 10	1.12	1.75	
10 – 12	1.50	1.75	
12 – 14	0.38	1.88	
14 – 16	0.87	1.50	
16 – 18	0.37	1.50	
18 – 20	0.88	1.26	
20 – 22	0.50	1.00	] PLATEAU 14 minutes
22 – 24	0.62	1.25	
24 – 26	0.75	1.01	
26 – 28	1.12	0.87	
28 – 30	1.37	0.87	← recovered
30 – 32	1.13	0.50	
PLATEAU TC1 during	t = 12 to 26 min = 14 minutes 60.50 °C → 69.25 °C		

The hot plate’s climb rate fell to **one third** of the control’s while passing through the paraffin melting range, then recovered and **exceeded** the control once melting was complete. **The control shows no corresponding feature at any temperature.**

9.2 The energy balance — the principal quantitative result

The Run A energy balance was integrated using the heat model **fitted to Run B** ($P = 7.26 \text{ W}$, $k = 0.1114 \text{ W/K}$), and the energy not accounted for by plate temperature rise was computed:

E_unaccounted(t) = ∫[P - k·(T_mean - T_amb)] dt - C_plates·ΔT_mean

Elapsed	Energy in (J)	Stored in plates (J)	Unaccounted (J)
10 min	3,907	2,458	1,449
18 min	6,383	3,924	2,459
26 min	8,521	4,787	3,735
30 min	9,470	5,606	3,864
34 min	10,287	6,426	3,861
38 min	10,987	7,116	3,871
45 min	11,989	8,064	3,924

MEASURED ABSORPTION	≈ 3,900 J	
THEORETICAL, 23.00 g	= 4,600 J	@ 200 J/g
ENERGY BALANCE CLOSES AT 85 %		

The unaccounted energy stops accumulating at approximately minute 30 — the same point at which the rate plateau of §9.1 ends. Two independent signals, computed by different methods from the same dataset, locate the completion of melting at the same time.

Why the layer’s sensible heat does not confound this. The model’s P and k were fitted to Run B, which contained 25.6 mL of liquid paraffin oil. The two layers have near-identical sensible heat capacity — oil 43.5 J/K, wax 48.3 J/K — so that term is largely common to both runs and cancels. The residual is dominated by latent heat.

A conservative floor by a second method. Comparing mean plate temperature directly against Run B and correcting for the 0.88 °C cooler start gives a peak deficit of 13.87 °C at 26 min, or **2,393 J**. This method ignores the lower heat loss of a cooler fixture and therefore understates the absorption.

MEASURED LATENT ABSORPTION: 2,393 J (floor) to 3,900 J (model)

A third signature. The pre-specified §6.2 power method, applied without modification, gives 7.73 W for Run C, 5.42 W for Run B and **2.77 W for Run A** — 36 % of the baseline. That method assumes all delivered power raises plate temperature. In Run A it does not. **The pre-registered analysis, applied blind, reports the energy as missing.**

9.3 Cast-wax contact resistance — a confound, separated

Identified during Run A at approximately four minutes, and reported in full because it was not anticipated in the pre-registration.

The wax was cast solid into the cavity. Paraffin contracts 10–15 % on solidifying, so the cast layer pulled away from the plate faces, leaving thin air voids. Air conducts at 0.026 W/m·K. Liquid paraffin oil, by contrast, wets both faces completely.

TC1 – TC2, early phase		
minutes	RUN A	RUN B
2	4.00	3.50
4	8.75	6.50
6	12.00	7.50
8	13.00	8.50
10	12.50	7.75
12	12.50	7.50

Run A’s temperature drop across the layer is **60 % larger than Run B’s before any melting is possible**. Paraffin melts at 58–62 °C; TC1 did not reach 58 °C until approximately minute 10.

This gap is thermal resistance, not latent heat, and it is not attributed to the mechanism. The two are separable:

BEFORE 10 min	elevated TC1–TC2 with NO rate anomaly → contact resistance
12 to 26 min	TC1 rate falls to one third of control while TC1 passes 60.5 → 69.3 °C → melting

The energy-balance result of §9.2 is unaffected. Contact resistance changes where the temperature sits, not how much energy the fixture holds. A resistance stores no energy.

Corrective action for Gen-0: the PCM must be melted in place in the assembled fixture and allowed to re-solidify under clamp load before the run begins. One thermal cycle removes the cast voids entirely.

9.4 An unexpected consequence, recorded because it was not predicted

23.00 g of paraffin at 0.90 g/cm³ occupies 25.6 cm³ against a 25.575 cm³ cavity. On melting, the liquid volume rises approximately 18 % above the cavity volume.

No wax was expelled. The tray and blanket beneath the fixture were inspected after cooling and were clean. **The expansion was absorbed by the cast voids of §9.3.** The defect and the expansion cancelled.

9.5 Why Run B's long delay is not principally the layer's own resistance

Two of Run B's failed predictions appear contradictory: the layer delayed heat far more than predicted while the temperature drop across it was far smaller than predicted.

The small ΔT follows from lower delivered power (7.73 W measured, not the 15 W assumed) and the foil bypass carrying 44 % of what crosses.

The long delay is that the layer **forces the hot plate to run hotter for the whole run**, and a hotter plate loses more heat to ambient. In Run C the plates sat 2.19 °C apart; in Run B, 7.38 °C apart for thirty-seven minutes, with the hot side shedding throughout.

The pre-registered model treated the layer as a passive resistance in a closed system. The measurement shows it also acts as a loss amplifier in an open one. That is a finding about the fixture, and it is more informative than a correct prediction would have been.

10. RESULTS AGAINST PREDICTIONS

Every quantitative prediction in §7.2 of the pre-registered report, scored.

#	Quantity	Predicted	Measured	
1	Delivered power	10 – 15 W	7.73 W	✗
2	$\Delta t_{\text{resistance}}$ (C→B)	100 – 250 s	923 s	✗
3	TC1 – TC2 steady, Run B	22 – 28 °C	7.38 °C	✗
4	Run B terminates early on TC1	likely	did not occur	✗
5	t_{85} Run A	17 – 27 min	> 45 min	✗
6	Plateau duration	13 – 25 min	14 min	✓
7	Plateau temperature	58 – 68 °C	60.5 – 69.3 °C	✓

Two further predictions behaved as the pre-registration itself anticipated: t_{85} for Runs C and B both exceeded their bands, and §7.2 states “predicted times are lower bounds... measured times are expected to exceed these figures.”

FIVE PREDICTIONS FAILED. TWO PASSED.

The five that failed are all predictions about the APPARATUS – delivered power, heat loss, thermal resistance, termination behaviour.

The two that passed are the two that test the MECHANISM – the duration and the temperature of the phase-change plateau.

The pre-registered model of the apparatus was substantially wrong. The pre-registered prediction of the mechanism was correct. Both are reported, and no prediction was revised after seeing data.

11. ERRORS FOUND AND CORRECTED DURING THE PROGRAMME

Recorded because they are part of the instrument's history and because a programme that reports no errors has not looked for them.

#	Error	How found	Consequence if missed
1	Crimped spade lugs in screw terminals caused intermittent open circuits	Faults correlated with cable movement	Random NaN rows throughout the dataset
2	−0.63 °C interpreted as instrument offset; +0.63 °C correction directed	Probe tips held together read −0.03 °C	Every reported difference displaced identically in all three runs — self-consistent and invisible
3	Plate heat capacity computed at 0.90 J/g·K (pure Al) instead of 0.88 (5052)	Recheck during report revision	2 % error in every energy figure
4	Fault-diagnosis current table wrong for series topology	Recomputed from measured resistance	Misdiagnosis of a genuine wiring fault
5	Foil described as “thermally invisible”	Recognised mid-run when ΔT stabilised far below prediction	44 % of crossing heat unaccounted
6	Insulation configuration inadequate	Heat-loss model fitted to Run C-1	Endpoint unreachable; whole programme void

12. LIMITATIONS

Stated in full, without qualification.

1. $n = 1$. One run per condition. No repeats, no error bars. Three t_{85} values cannot be distinguished from three draws from an unknown distribution, and this report does not claim otherwise.
2. SUBSTITUTE MATERIAL. Paraffin wax, not RT64HC. Different melt point, different latent heat, different chemistry.
3. LATENT HEAT ASSUMED, NOT MEASURED. 200 J/g is a literature value. Commercial paraffin ranges 180–220 J/g; against that band the energy balance closes at 77 % to 94 %. No DSC trace of the actual wax exists.
4. ONE LAYER OF FIVE. Nothing tested about pyrolytic graphite, expandable graphite, gas capture, or dielectric isolation.
5. NO CELL. Nothing entered thermal runaway. This is a resistive heater and two aluminium plates.
6. FOIL BYPASS carries 44 % of crossing heat, present in Runs B and A alike. Biases results conservatively.
7. CAST-WAX CONTACT RESISTANCE present in Run A, absent in Run B. Quantified and separated, but not eliminated.
8. PRIMARY ENDPOINT NOT REACHED in Run A. $t_{85}(A)$ is a bound.
9. ABSOLUTE TEMPERATURES carry the ± 2.5 °C K-type Class 2 tolerance. Differences between runs do not, as any systematic offset is common to all runs and cancels.
10. WAX CHARGE of 23.00 g carries a small volume inconsistency (§9.4). At 21.75 g the balance closes at 90 % instead of 85 %. The conclusion is robust across the range.
11. CLOCK TIMES for Run A reconstructed from timestamped screenshots. Elapsed times are exact and govern.
12. SINGLE OPERATOR. No independent replication, no second observer, no blinding.

13. CONCLUSIONS

1. The mechanism is real and measurable. A phase-change layer absorbed approximately 3,900 J — 85 % of the theoretical heat of fusion of the charge — and suppressed the heated plate’s temperature rise for 14 minutes, inside both predicted bands. Three independent methods agree.

2. H1 is not falsified. $\Delta t_{\text{latent}} > 455$ s against a falsification threshold of 60 s, a margin of 7.6 $\times$, reported as a lower bound by a method published before the run.

3. The quantitative model of the apparatus was poor. Five of seven predictions failed, several by factors of 3–4 $\times$. Delivered power, heat loss and thermal resistance were all substantially mispredicted. This is a real finding about the current state of the modelling and is not minimised here.

4. The result is conservative. The foil bypass and the reported lower bound both understate the mechanism. The programme’s two most significant apparatus artefacts bias against the hypothesis.

5. Nothing here is evidence about the product. One layer, a substitute material, no cell, no runaway, n = 1.

14. NEXT STAGE — GEN-0

PRE-GEN-0	← completed by this report One layer, bench scale, aluminium plates and cartridge heaters.
GEN-0	The first real five-layer pad, fabricated to the build manual. Real PGS, RT64HC, expandable graphite, zeolite/carbon and PVDF, laminated into one 1.94 mm pad. Question: does one real cell entering thermal runaway propagate to its neighbour across it? Output: CELL-T0-CELL PROPAGATION DATA.
GEN-1	Module level, GB 38031-2025-style propagation test.
GEN-2	Full pack, third-party certification against GB 38031-2025.

Three method changes carried forward from this programme:

1.	Melt the PCM in place before the run. Removes cast voids and the contact resistance of \$9.3.
2.	Solve containment before the programme starts. The foil was a sound field solution but it permanently contaminated the comparison with a 0.308 W/K bypass.
3.	n = 3 minimum per condition. Every other element of this method is rigorous and then rests on a single run.

15. DATA AVAILABILITY

All raw data is published unedited. No row was removed, smoothed or corrected.

run_C_ABORTED_20260826.csv	Run C-1, aborted, 12.4 min
run_C2_20260826.csv	Run C-2, baseline, 1,600 samples
run_B_20260827.csv	Run B, control, 2,501 samples
run_A_20260829.csv	Run A, test, 3,356 samples
gate2_sensing_check_20260824.csv	Gate 2 still-air, 836 samples

Repository: github.com/Idreesh111/btsb-bench-test-preregistration

15.1 Document register

CIP-BTSB-BT-001 Rev A	pre-registered protocol	26 Aug
Record G1	software chain	25 Aug
Record G2	sensing (rev. 25 Aug)	24 Aug
Record G3	heater circuit	25 Aug
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Record R-A	Run A	29 Aug
THIS REPORT	consolidated final	29 Aug

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